30/09/2026

Code-B Phase-2

Aakash
Medical|IIT-JEE|Foundations
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MM : 720 **Fortnightly Test for NEET-2027 (XII Passed) Ph-2 - FT05B** **Time : 180 Min.**

Topics Covered:
Physics : Thermal Properties of Matter, Thermodynamics, Kinetic Theory
Chemistry : Equilibrium
Botany : Anatomy of Flowering Plants
Zoology : Locomotion & Movement-II: (Upto muscles and bone disorders), Neural Control & Coordination

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General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

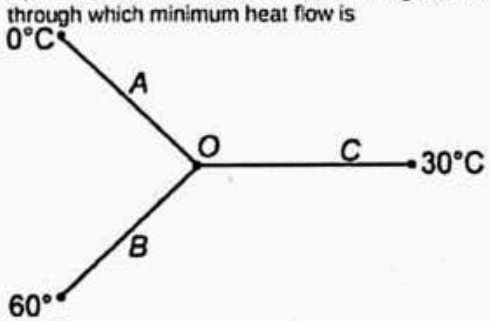
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PHYSICS

1. Temperature is a measure of
   (1) Hotness and coldness
   (2) Heat possessed by a body
   (3) Potential energy of body
   (4) Both (2) and (3)
2. In a thermometer the ice point (lower fixed point) is marked as 10° and the steam point (upper fixed point) is marked as 130°. At 40°C temperature, what will this thermometer read.
   (1) 28° (2) 39°
   (3) 58° (4) 40°
3. **Statement A :** Two bodies are said to be in thermal equilibrium when no (net) heat flows from one body to other.
   Statement B: At thermal equilibrium, two bodies can be at different temperature.
   (1) Statement A is true but statement B is false
   (2) Statement A is false but statement B is true
   (3) Both statements A and B are true
   (4) Both statements A and B are false
4. Time period of a pendulum is 4 s and coefficient of linear expansion of material of its string is $2 \times 10^{-3}\ K^{-1}$. If temperature is increased by 50°C, then change in time per second is
   (1) 0.2 s (2) 0.02 s
   (3) 0.05 s (4) 0.1 s
5. The relation between coefficient of linear expansion (alpha) and coefficient of volume expansion (gamma) for an isotropic solid is
   (1) $\gamma = 2\alpha$ (2) $3\gamma = \alpha$
   (3) $\gamma = 3\alpha$ (4) $2\gamma = 3\alpha$
6. A solid sphere and a hollow sphere of same size and of same metal are heated to same temperature difference, then
   (1) Both spheres will expand equally
   (2) Solid sphere will expand more than hollow sphere
   (3) Hollow sphere will expand more than solid sphere
   (4) Solid sphere will expand but hollow sphere does not
7. Heat required to convert 10 g of ice at –10°C to water at 20°C is
   (1) 250 cal (2) 2916 cal
   (3) 850 cal (4) 1050 cal
8. The ratio of radii of two solid spheres made of same material is 1 : 3. The ratio of their heat capacity will be
   (1) 1 : 1 (2) 1 : 27
   (3) 3 : 1 (4) 9 : 1
9. On centigrade scale the temperature of body increases by 45 degrees. The corresponding increase in temperature on kelvin scale is
   (1) 318 K (2) 45 K
   (3) 245 K (4) 363 K
10. A body cools in 10 minutes from 60 °C to 40 °C. The reducing of temperature of body in next 10 minutes from 40 °C is (temperature of surrounding is 20 °C)
    (1) Greater than 20 °C (2) Less than 20 °C
    (3) Equal to 20 °C (4) Equal to 40 °C

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11. Three identical conductors of same material and dimension A, B and C are fixed at *O* as shown in figure. The conductor through which minimum heat flow is



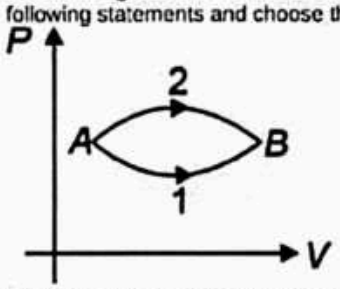
(1) A
(2) C
(3) B
(4) All have same heat of flow

12. Match the following for work done in various thermodynamic processes. (Symbols have their usual meaning)

List-I (Process)	List-II (W.D.)
A. Isobaric expansion	(I) Zero
B. Isochoric	(II) $-\Delta U$
C. Adiabatic	(III) ΔQ
D. Isothermal	(IV) $nR\Delta T$

(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-I, C-III, D-II
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-II, C-III, D-IV

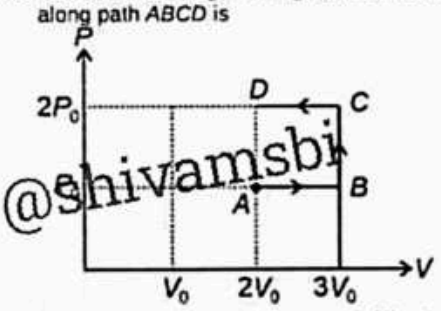
13. Two paths for the change of state of a gas from *A* to *B* is shown in figure. On the basis of given graph, consider the following statements and choose the correct option



(a) Work done in path (2) is increasing
(b) Work done in path (1) is decreasing
(c) Work done in path (1) is increasing
(d) Work done in path (2) is decreasing
(1) Only (a) and (c) are correct
(2) Only (b) and (c) are correct
(3) Only (b) and (d) are correct
(4) Only (a) and (d) are correct

14. The ratio of internal energy of molecules of neon and hydrogen at temperature 127 °C is [Assume both the gases to be ideal]
(1) 3/5
(2) 1
(3) 2/5
(4) 2

15. Consider following *PV*-diagram. Work done by an ideal gas along path *ABCD* is



(1) P_0V_0
(2) $-P_0V_0$
(3) $2P_0V_0$
(4) $4P_0V_0$

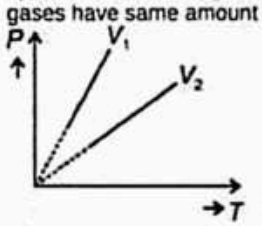
16. One mole of an ideal gas expands according to the relation $\frac{P}{\sqrt{V}}$ = Constant. When the temperature of gas changes by 30°C, the work done by gas will be (Where symbols have their usual meaning)
(1) $10R$
(2) $20R$
(3) $30R$
(4) $40R$

17. A : Change in internal energy in a closed path may be positive or may be negative but not zero.
R : Change in internal energy is path dependent quantity.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

18. The change in internal energy of a gaseous system, if heat supplied to the gaseous system is 150 J and work done by the system is 80 J will be
(1) 70 J
(2) 80 J
(3) 100 J
(4) Zero

19. Specific heat of a gas undergoing isothermal change is
(1) Zero
(2) Positive
(3) Infinite
(4) Negative

20. In the following $P - T$ diagram which among the given options is correct regarding volume of gas? (Assume both gases have same amount of moles)



(1) $V_1 > V_2$
(2) $V_1 < V_2$
(3) $V_1 = V_2$
(4) Relation between V_1 and V_2 can't be predicted

21. Temperature of one mole of an ideal gas is raised by 2°C by a process. The internal energy change would be maximum if the process is
(1) Isochoric
(2) Adiabatic
(3) Isobaric
(4) Equal for all the processes

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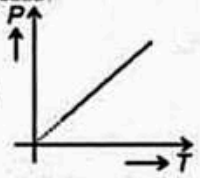
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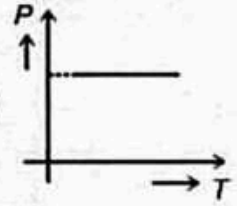
22. In adiabatic change, the pressure and temperature of a linear triatomic gas are related as $P \propto T^k$, where k equals (ignore vibrational modes)

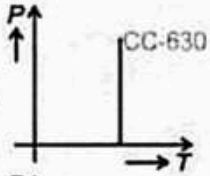
(1) $\frac{5}{2}$ (2) $\frac{1}{3}$

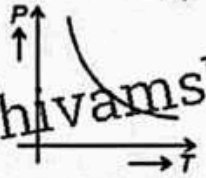
(3) $\frac{7}{2}$ (4) $\frac{7}{3}$

23. Which of the following graphs corresponds to isochoric process?

(1) 

(2) 

(3) 

(4) 

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24. Mean translational kinetic energy per molecule for diatomic ideal gas is (k_B : Boltzmann constant)

(1) $\frac{3}{2}RT$ (2) $\frac{3}{2}k_BT$

(3) $\frac{5}{2}RT$ (4) $\frac{1}{2}k_BT$

25. Efficiency of a heat engine working between 127°C and 327°C may be

(1) 20% (2) 25%

(3) 33% (4) All of these

26. Consider the following statements and choose the **incorrect** option

(1) According to second law of thermodynamics, no engine can have 100% efficiency

(2) Irreversible engines have more efficiency than reversible engines which consist of roversible quasi-static processes

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(3) A Carnot engine is an ideal reversible engine that works between two reservoirs at different temperatures

(4) All of these

27. **Assertion (A) :** In an adiabatic expansion process, negative change in internal energy of a gas is equal to workdone by the gas.

Reason (R) : Temperature of gas remains constant in free expansion.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

28. The carnot cycle consist of

(1) Two irreversible isothermal process and two irreversible adiabatic process

(2) Two irreversible isothermal process and two reversible adiabatic process

(3) Two reversible isothermal process and two reversible adiabatic process

(4) Two reversible isothermal process and two irreversible adiabatic process

29. **Statement A:** The value of $\frac{C_V}{C_P}$ for a monoatomic gas will be less as compared to that of a diatomic gas. (Symbols have their usual meanings)

Statement B: The molecules of a diatomic gas have more degree of freedom than those of monoatomic gas.

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(1) Both statement (A) and statement (B) are correct

(2) Both statement (A) and statement (B) are incorrect

(3) Statement (A) is correct and statement (B) is incorrect

(4) Statement (A) is incorrect and statement (B) is correct

30. For an ideal gas, the r.m.s speed at 800K is

(1) Half the value at 200K

(2) Double the value at 200 K

(3) Same as at 200 K

(4) Four times the value at 200 K

31. If pressure of a gas is doubled at constant temperature then mean free path of gas molecules will be

(1) Thrice (2) Twice

(3) Half (4) Same

32. An ideal monoatomic gas is compressed suddenly to $\left(\frac{1}{8}\right)^{th}$ of its initial volume. If initial pressure of the gas is P_0, then its final pressure is

(1) $8P_0$ (2) $32P_0$

(3) $16P_0$ (4) $4P_0$

33. Which among the following gas has only translational degree of freedom?

(1) Argon (2) Oxygen

(3) Nitrogen (4) Carbon dioxide

34. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

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(1) $4RT$ (2) $15RT$

(3) $9RT$ (4) $11RT$

35. When temperature of an ideal gas contained in closed vessel is increased by 4°C, its pressure increases by 0.5%. The original temperature of the gas was approximately

(1) $527^\circ C$ (2) $800^\circ C$

(3) $627^\circ C$ (4) 527 K

36. Which of the following possesses maximum most probable speed? (All are at the same temperature)

(1) Oxygen (2) Hydrogen

(3) Carbon dioxide (4) Helium

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37. The ratio of the density and pressure of a fixed mass of an ideal gas is "5" at 10°C. Then this ratio at 110°C is
(1) 3.69 (2) 5
(3) 8.46 (4) 6.76

38. The slope of P-V graph for isochoric process is
(1) Zero (2) Infinite
(3) One (4) Two

39. **Assertion (A) :** Eskimos make double wall houses of ice blocks
Reason (R) : The air trapped between double walls prevents the conduction of heat energy from inside the house to outside it.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

40. Match List I with List II

	Physical quantity		Unit
(a)	Coefficient of linear expansion (α)	(i)	$J\ K^{-1}$
(b)	Heat capacity (C)	(ii)	K^{-1}
(c)	Specific heat capacity (c)	(iii)	$J\ kg^{-1}$
(d)	Latent heat (L)	(iv)	$J\ kg^{-1}\ K^{-1}$

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

41. Select the correct statement(s) regarding heat transfer
(1) Natural convection requires gravity
(2) Forced convection can occur in both gravity and gravity free space
(3) Radiation occurs in space only where gravity is present
(4) Both (1) & (2)

42. In the range 273 K to 277 K, the correct statement(s) is/are
(1) Water expands on heating
(2) Water contracts on cooling
(3) Maximum density of water occurs at 277 K
(4) All of these

43. The unit of thermal conductivity is
(1) $J\ m^{-1}K^{-1}$ (2) $J\ m^{-2}K^{-1}$
(3) $W\ m^{-1}K^{-1}$ (4) $W\ m\ K^{-1}$

44. The wavelength corresponding to maximum intensity of radiation from a black body at temperature 4000 K is 5000 Å. If temperature of body is decreased by 2000 K, then emitted wavelength corresponding to maximum intensity will be
(1) 8000 Å (2) 6000 Å
(3) 4000 Å (4) 10000 Å

45. Consider the statements regarding the mean free path of gas molecules.
(a) It is inversely proportional to number density of gas molecules.
(b) It is inversely proportional to pressure of gas.
(c) It is inversely proportional to square of diameter of gas molecule.
Based on above information pick the correct option.
(1) Only (a) and (b) are correct
(2) Only (a) is correct
(3) Only (a) and (c) are correct
(4) All (a), (b) and (c) are correct

CHEMISTRY

46. Which among the following statement(s) is incorrect for physical equilibrium?
(1) Equilibrium is possible only in a closed system at a given temperature
(2) Both the opposing processes occur at the same rate and there is a static but stable condition
(3) All measurable properties of the system remain constant
(4) When equilibrium is attained for a physical process, it is characterised by constant value of one of its parameters at a given temperature

47. Which of the following will change the value of equilibrium constant?
(1) Change in pressure
(2) Change in concentration of product
(3) Change in amount of catalyst
(4) Change in temperature

48. Consider the following reactions at equilibrium and their respective equilibrium constants:
$N_2 + 3H_2 \rightleftharpoons 2NH_3\ K_1$
$N_2 + O_2 \rightleftharpoons 2NO\ K_2$
$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O\ K_3$
The equilibrium constant (K) for the following reaction will be
$2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$
(1) $\frac{K_1}{K_3^3 K_2}$ (2) $K_1 K_2 K_3^3$
(3) $\frac{K_2 K_3^3}{K_1}$ (4) $\frac{K_1 K_3^3}{K_2}$

49. Ammonia dissociates as follows, in a closed vessel, $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$. If total pressure of the mixture at equilibrium is P_T and degree of dissociation of NH_3 is x, then equilibrium constant will be numerically equal to (assuming degree of dissociation is negligibly small)
(1) $3x^4P_T^2$ (2) $\frac{27x^4P_T^2}{16}$
(3) $\frac{9x^4P_T^2}{8}$ (4) $\frac{9x^2P_T^2}{8}$

50. An example of Heterogeneous equilibrium is,
(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
(3) $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$
(4) $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$

51. Two solids dissociate in a container as follows:
A(s) $\rightleftharpoons$ B(g) + C(g); K_{p_1} = a atm^2
D(s) $\rightleftharpoons$ C(g) + E(g); K_{p_2} = b atm^2
The total equilibrium pressure when both the solids dissociate simultaneously is:
(1) (a + b) atm (2) $2\left(\sqrt{a+b}\right)$ atm
(3) $(a + b)^2$ atm (4) $\frac{a+b}{\sqrt{2}}$ atm

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52. Given below are the two statements:
Statement I: If $\Delta G^\circ < 0$, then $K > 1$.
Statement II: If ΔG of a reaction is negative, then the reaction is spontaneous and proceeds in forward direction.
In light of above statements, choose the correct option.
(1) Statement I is correct but statement II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct

53. Consider the given reaction
$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
If degree of dissociation of 1 mol $SO_3(g)$ to $SO_2(g)$ and $O_2(g)$ is 20%, then value of K_c will be (volume of container = 1 L)
(1) 6.25×10^{-3} (2) 6.25×10^{-2}
(3) 2.25×10^{-3} (4) 2.25×10^{-2}

54. Methanol can be prepared as follows
$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$;
$K_P = 3 \times 10^{-4}$ at 227°C
Total pressure required to convert 25% of CO to CH_3OH at 227°C, if CO and H_2 are obtained from following reaction, will be
$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$
(1) 25.12 bar (2) 46.67 bar
(3) 32.25 bar (4) 19.28 bar

55. The reaction quotient (Q) at equilibrium is always
(1) Equal to one (2) Equal to 'K_{eq}'
(3) Greater than 'K_{eq}' (4) Less than 'K_{eq}'

56. Select the correct expression
(1) $\Delta G = -\Delta G^\circ + RT \ln K$ (2) $\Delta G = \Delta G^\circ - RT \ln K$
(3) $\Delta G = \Delta G^\circ + RT \ln Q$ (4) $\Delta G^\circ = RT \ln K$

57. Equilibrium constant for $HI(g) \rightleftharpoons \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g)$ is 8. Equilibrium constant for $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ will be
(1) 64 (2) $\frac{1}{64}$
(3) $\frac{1}{8}$ (4) $\frac{1}{16}$

58. At 300 K, $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ has an equilibrium constant, $K_c = 4.0 \times 10^{31}$. At equilibrium, this reaction has
(1) Very high concentration of HCl as compared to H_2
(2) Very low concentration of HCl as compared to Cl_2
(3) Appreciable concentrations of Cl_2 as well as HCl
(4) Appreciable concentrations of H_2 as well as HCl

59. Consider the following statements.
Statement I: The value of equilibrium constant is independent of initial concentrations of the reactants and products
Statement II: If $K_c = 10^{-3}$, products predominates over reactants.
In the light of above statements choose the correct option.
(1) Statement I is incorrect but statement II is correct
(2) Statement I is correct but statement II is incorrect
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

60. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): Introduction of inert gas at constant volume and temperature does not affect the equilibrium state of a gaseous reaction.
Reason (R): Partial pressure of reacting gases is unchanged on introduction of inert gas at constant volume and temperature.
In the light of the above statements, choose the correct answer from the options given below.
(1) Both **(A)** and **(R)** are true but **(R)** is **NOT** the correct explanation of **(A)**
(2) **(A)** is true but **(R)** is false
(3) **(A)** is false but **(R)** is true
(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

61. Consider the following statements about catalyst
(a) It lowers the activation energy for the forward and reverse reactions by exactly the same amount
(b) It increases the equilibrium constant value of the reaction
(c) it does not appear in the equilibrium constant expression
The correct statements are
(1) (a) and (b) only (2) (a) and (c) only
(3) (a), (b) and (c) (4) (b) and (c) only

62. If at equilibrium HI dissociates 50% at 25°C, then ΔG° for the reaction, $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ is
(1) 1.386 RT (2) –2.303 RT
(3) 0.693 RT (4) –0.693 RT

63. Which one of the following condition will favour maximum formation of the product in the reaction at equilibrium
$A_2(g) + 2B_2(g) \rightleftharpoons C(g), \Delta_r H = +50\ kJ\ mol^{-1}$
(1) High temperature (2) Low pressure
(3) Adding more A_2 (4) Adding more C

64. Match List I with List II

	List-I (Reactions)		List-II (K_c/K_p)
a.	$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$	(i)	1
b.	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(ii)	RT
c.	$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$	(iii)	$\frac{1}{(RT)^2}$
d.	$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$	(iv)	$\frac{1}{RT}$

Choose the **correct** option
(1) a-(ii), b-(i), c-(ii), d-(iv) (2) a-(iii), b-(i), c-(iv), d-(ii)
(3) a-(iv), b-(iii), c-(ii), d-(i) (4) a-(iv), b-(ii), c-(i), d-(iii)

65. Given below are two statements
Statement-I: The increase of pressure on ice(s) $\rightleftharpoons$ water (l) system at constant temperature will lead to forward shift of equilibrium.
Statement-II: The decrease of pressure on $H_2O(l) \rightleftharpoons H_2O(g)$ system at constant temperature will lead to the backward shift of equilibrium.
In the light of above statements, choose the correct answer from the options given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

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66. Given below are two statements
Statement I: K_C increases with increase in temperature for endothermic reaction.
Statement II: K_C decreases with increase in temperature for exothermic reaction.
In the light of above statements, choose the most appropriate answer from the options given below
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

67. Conjugate base of HPO_4^{2-} is
(1) PO_4^{3-} (2) $H_2PO_4^-$
(3) H_3PO_4 (4) H_4PO_3

68. Degree of dissociation of 0.1 M acetic acid with $K_a = 1.8 \times 10^{-5}$ is
(1) 13.4% (2) 1.34%
(3) 5.3% (4) 4.2%

69. The pH of 10^{-8} M NaOH solution is
(1) 6.98 (2) 6
(3) 7.02 (4) 8

70. If 100 mL solution of pH = 2 is mixed with 100 mL of solution of pH = 3, then pH of mixture will be
(1) 1.5 (2) 2.26
(3) 0.39 (4) 3.5

71. Which among the following is a Lewis acid?
(1) H_2O (2) B_2H_6
(3) NH_3 (4) CH_3OH

72. Consider the following statements
(a) HF is a stronger acid than H_2O
(b) $AlCl_3$ is a Lewis acid
(c) K_w of water at 298 K is 1×10^{-7}
The correct statements are
(1) (b) and (c) only (2) (a) and (b) only
(3) (a), (b) and (c) (4) (a) and (c) only

73. The rate constants of forward and backward reactions are 8.5×10^{-5} and 1.7×10^{-4} respectively. The equilibrium constant is
(1) 0.5 (2) 50
(3) 0.02 (4) 0.2

74. The salt with highest pH is
(1) Sodium acetate (2) Sodium chloride
(3) Sodium sulphate (4) Ammonium acetate

75. pH of 0.04 M Calcium acetate solution is (pk_a of CH_3COOH = 4.74)
(1) 4.42 (2) 10.25
(3) 8.82 (4) 7.25

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76. Match List I with List II.

	List I (Mixture of acid and base)		List II (pH of mixture at 25°C)
a.	100 ml, 0.1M NaOH + 50 ml, 0.1M H_2SO_4	(i)	$pH = 7 + \frac{1}{2}(pK_a + \log c)$
b.	100 ml, 0.1M NaOH + 100 ml, 0.1M CH_3COOH	(ii)	$pH = 7 + \frac{1}{2}(pK_a - pK_b)$
c.	100 ml, 0.1M NH_4OH + 100 ml, 0.1M CH_3COOH	(iii)	pH = 7
d.	100 ml, 0.1M HCl + 100 ml, 0.1M NH_4OH	(iv)	$pH = 7 - \frac{1}{2}(pK_b + \log c)$

Choose the correct option.
(1) a-(i), b-(ii), c-(iii), d-(iv) (2) a-(iv), b-(iii), c-(ii), d-(i)
(3) a-(iii), b-(i), c-(ii), d-(iv) (4) a-(iii), b-(ii), c-(iv), d-(i)

77. 100 mL, 0.1 M CH_3COOH ($K_a = 10^{-5}$) is titrated against 0.05 M solution of NaOH.
a. At the start of titration, pH of solution is 3.
b. If 100 mL of NaOH is added, pH of solution will be 5.
c. If 200 mL of NaOH is added, pH of solution will be 8.76
The correct statement(s) is/are
[Given : log 3 = 0.477]
(1) a, b and c (2) b and c only
(3) a and b only (4) c only

78. The percentage ionisation of 0.02 M acetic acid in 0.2 M HCl will be nearly (K_a of acetic acid is 1.74×10^{-5})
(1) 0.009% (2) 0.08%
(3) 0.002% (4) 0.02%

79. Match the following columns :

Column I (Salt)	Column II (Type of Hydrolysis)
a. Ammonium chloride	(i) Cationic hydrolysis
b. Sodium acetate	(ii) No hydrolysis
c. Ammonium acetate	(iii) Anionic hydrolysis
d. Sodium chloride	(iv) Both cationic and anionic hydrolysis

Choose the option containing most appropriate match.
(1) a(i), b(iv), c(iii), d(ii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(ii), b(iv), c(i), d(iii) (4) a(i), b(iii), c(iv), d(ii)

80. The solubility of AgCl will be minimum in
(1) 0.15 M NH_3 (2) 0.1 M $CaCl_2$
(3) 0.15 M $AgNO_3$ (4) Pure water

81. pH of saturated solution of $Ca(OH)_2$ is (K_{sp} of $Ca(OH)_2$ = 4 $\times 10^{-6}$)
(1) 12.3 (2) 11.9
(3) 10.7 (4) 13.3

82. The molar solubility of AgCl in 0.1 M solution of NaCl is (K_{sp} of AgCl = 1.8×10^{-10})
(1) 1.8×10^{-8} M (2) 1.8×10^{-9} M
(3) 1.34×10^{-5} M (4) 1.8×10^{-5} M

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83. Match the List I and List II.

	List I (Salt)		List II (solubility in terms of solubility product K_{sp})
a.	$BaSO_4$	(i)	$\left(\frac{K_{sp}}{108}\right)^{1/5}$
b.	Hg_2Cl_2	(ii)	$\left(\frac{K_{sp}}{27}\right)^{1/4}$
c.	$Ca_3(PO_4)_2$	(iii)	$\left(\frac{K_{sp}}{4}\right)^{1/3}$
d.	$Fe(OH)_3$	(iv)	$(K_{sp})^{1/2}$

The correct match is
(1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(i), c(iv), d(ii)

84. At 85°C, distilled water has $[H_3O^+]$ concentration equal to 1×10^{-6} mole/litre. The value of K_W at this temperature will be
(1) 1×10^{-8} (2) 1×10^{-14}
(3) 1×10^{-12} (4) 1×10^{-7}

85. In closed system : $P(s) \rightleftharpoons 2Q(g) + 3R(g)$
if the partial pressure of R is tripled then partial pressure of Q will become
(1) Half of its original pressure
(2) $\frac{1}{3\sqrt{3}}$ times the original pressure
(3) $2\sqrt{2}$ times its original pressure
(4) Twice of the original pressure

86. Strong electrolytes have their degree of dissociation (α) values nearly
(1) 0.1 (2) 0.2
(3) 0.5 (4) 1

87. Solubility products of the salts of type MX, MX_2, M_3X at temperature T are 4×10^{-8}, 3.2×10^{-14} and 27×10^{-16} respectvely. Solubility ($moldm^{-3}$) of the salts at temperature T are in the order
(1) $MX > MX_2 > M_3X$ (2) $M_3X > MX_2 > MX$
(3) $MX_2 > M_3X > MX$ (4) $MX > M_3X > MX_2$

88. The ratio of concentration of salt to base is 10 : 1 for a basic buffer. If the pK_b of base is 5, then the pOH of buffer solution is
(1) 5 (2) 6
(3) 8 (4) 9

89. Addition of inert gas at constant pressure has same effect as of
(1) Increasing temperature (2) Decreasing temperature
(3) Increasing volume (4) Increasing reactants

90. Given below are two statements.
Statement I: Dissociation of aq HCN decreases on addition of aq KCN solution.
Statement II: Common ion effect is a phenomenon based on the Le Chatelier's principle.
In the light of above statements, choose the correct option.
(1) Both the statement I and statement II are correct
(2) Both the statement I and statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

BOTANY

91. Match **List-I** with **List-II**

List-I	List-II
a. Dicot stem	(i) Conjoint, closed vascular bundles
b. Dicot root	(ii) Hypodermis is sclerenchymatous
c. Monocot leaf	(iii) Semi-lunar shaped patches of pericycle
d. Monocot stem	(iv) Radial, exarch type of vascular bundles

Choose the **correct** option
(1) a(iv), b(ii), c(i), d(iii) (2) a(i), b(ii), c(iv), d(iii)
(3) a(iii), b(iv), c(i), d(ii) (4) a(ii), b(i), c(iv), d(iii)

92. Read the given assertion (A) and reason (R) and choose the **correct** option.
Assertion (A) : In stems of grasses, large sized vascular bundles are arranged in a ring-like manner.
Reason (R) : The large sized vascular bundles in monocot stem are densely arranged towards the periphery as compared to smaller ones.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

93. A student observed V.S. of some plant structure having following features.
(a) Conjoint but closed vascular bundles.
(b) Epidermis on upper and lower surfaces with conspicuous cuticle.
(c) Chloroplast containing parenchymatous tissue divided into two regions.
(d) Stomata mostly on abaxial surface.
Identify the plant structure based on the given features.
(1) Monocot stem (2) Dicot root
(3) Dicot leaf (4) Monocot leaf

94. Identify the statements as **true (T)** or **false (F)** and choose the **correct** option.
A. Conjunctive tissue is parenchyma cells which lie between xylem and phloem in dicot root.
B. The conjoint vascular bundles usually have the phloem located only on the outer side of xylem..

	A	B
(1)	T	T
(2)	T	F
(3)	F	T
(4)	F	F

(1) (1) (2) (2)
(3) (3) (4) (4)

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95. Fill the position of A & B with appropriate option.

Character	Monocot root	Monocot stem
Epidermis	Parenchymatous	Parenchymatous
Endodermis	Parenchymatous	A
Pericycle	B	Absent

	A	B
(1)	Sclerenchymatous	Absent
(2)	Absent	Collenchymatous
(3)	Parenchymatous	Absent
(4)	Absent	Parenchymatous

(1) (1) (2) (2)
(3) (3) (4) (4)

96. Which of the following statements are **correct** w.r.t. isobilateral leaf?
a. Stomata are almost equally distributed on both surface.
b. Mesophyll has no differentiation as palisade and spongy parenchyma.
c. Bulliform cells may be present in some cases in the upper epidermis.
d. Only dorsal surface is cuticularized.
(1) Only a and b (2) Only a, b and c
(3) Only a, c and d (4) All of these

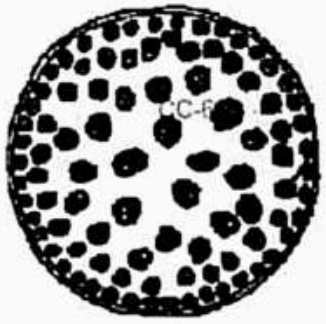
97. The central most portion of stem of dicotyledonous plants is occupied by
(1) Vascular bundles (2) Pericycle
(3) Pith (4) Cortex

98. Identify the **incorrect** statement regarding root endodermis.
(1) Its cells do not have intercellular spaces.
(2) It is a single layered structure.
(3) Its cells have deposition of waxy material lignin in the form of casparian strips.
(4) It consists of barrel-shaped cells.

99. All of the following elements of complex tissues are present in monocots, **except**
(1) Xylem parenchyma (2) Phloem parenchyma
(3) Xylem fibres (4) Phloem fibres

100. Pericycle of monocot roots
(1) Forms lateral roots
(2) Dedifferentiates and forms a part of vascular cambium
(3) Is the innermost layer of stele
(4) Develops at the time of secondary growth

101. The figure depicted below is transverse section of



(1) Monocot stem (2) Monocot root
(3) Dicot stem (4) Dicot root

102. All are related to guard cells, **except**
(1) In dicots, inner walls are thick
(2) These are sclerenchymatous
(3) Contain chloroplast
(4) Regulate opening and closing of stomata

103. The type of vascular bundle in which xylem and phloem are situated at same radius, is commonly present in
(i) Root
(ii) Stem
(iii) Leaf
(1) (i) & (ii) (2) (ii) & (iii)
(3) (i) & (iii) (4) Only (i)

104. Select the **incorrect** statement w.r.t. monocot root
(1) Pith is small and inconspicuous
(2) Cambium is absent
(3) Vascular bundles is polyarch
(4) There is no secondary growth

105. Epiblema is
(1) Innermost layer of the cortex
(2) Outermost layer of stele in monocot
(3) Outermost epidermal layer in roots
(4) Storage tissue in cortex

106. Read the following statements and select the **correct** option.
Statement A : Bulliform cells in the leaves absorb water and due to absorption of water, they become turgid.
Statement B : During insufficient water supply, bulliform cells lose the water and become flaccid due to water loss.
(1) Both statement A and B are correct
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both statements A and B are incorrect

107. Read the following statements and select the **correct** option.
A. Maize stem cannot be distinguished from gram stem on the basis of relative positions of protoxylem and metaxylem.
B. Phloem parenchyma and water containing cavities are generally found in vascular bundles of monocot stems.
(1) Only A is correct (2) Only B is correct
(3) Both A & B are correct (4) Both A & B are incorrect

108. Root hair and trichome both are
(1) Always unicellular
(2) Branched
(3) Epidermal outgrowths
(4) Responsible for preventing water loss

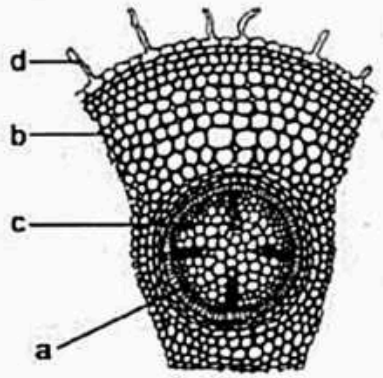
109. In roots, the vascular bundles are
(1) Radial & endarch (2) Radial & exarch
(3) Conjoint & endarch (4) Conjoint & exarch

110. Trichome is a soft or stiff hairs, present on the stem of a plant and it cannot be an organ of
(1) Protection (2) Secretion
(3) Reproduction (4) Preventing water loss

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111. Mark a, b, c and d



(1) Metaxylem, protoxylem, cortex, trichome
(2) Endodermis, cortex, protoxylem, root hair
(3) Endodermis, cortex, metaxylem, root hair
(4) Pericycle, cortex, metaxylem, trichome

112. Stele includes
(1) Vascular tissue, cortex and pith
(2) Vascular tissue and pith only
(3) Vascular tissue, pith and pericycle
(4) Vascular tissue, pith, pericycle and endodermis

113. The vascular bundle is said to be open when
(1) Phloem is not covered by xylem
(2) Cambium is present between xylem and phloem
(3) Xylem and phloem are situated at different radii
(4) They are not covered by bundle sheath

114. The specialised epidermal cells, present in the vicinity of guard cells are called
(1) Bulliform cells (2) Companion cells
(3) Subsidiary cells (4) Endodermal cells

115. Bundle sheath and conjoint-closed vascular bundles are present in all given parts of plants, **except**
(1) Stem of dicotyledonous plants
(2) Stem of grasses
(3) Dorsiventral leaves
(4) Isobilateral leaves

116. How many xylem and phloem patches are usually present in dicot root?
(1) Two to four (2) Eight to ten
(3) Five to eight (4) More than ten

117. In between the vascular bundles of dicot stem, there are a few layers of radially placed parenchymatous cells, which constitute
(1) Bundle sheath (2) Medullary rays
(3) Cambium (4) Pericycle

118. The primary adaptive advantage provided by bulliform cells in grasses during water stress is that they
(1) Enhance nutrient uptake from the soil.
(2) Increase the rate of photosynthesis by expanding the leaf surface.
(3) Reduce transpiration by causing leaf rolling to minimize exposed surface area.
(4) Promote rapid water transport to the leaf tips during dehydration

119. The innermost layer of cortex is called
(1) Epidermis (2) Endodermis
(3) Hypodermis (4) Pericycle

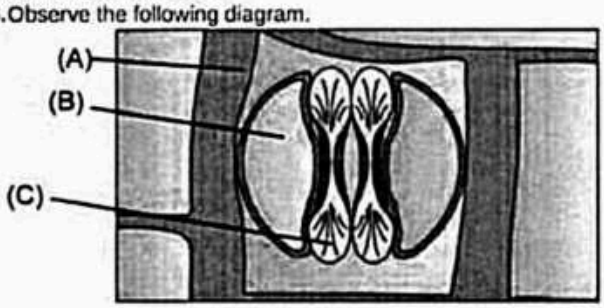
120. In dicot stem, hypodermis is composed of
(1) Sclerenchyma (2) Collenchyma
(3) Parenchyma (4) Casparian strips

121. Ground tissue of leaves is called
(1) Stele (2) Cambium
(3) Mesophyll (4) Vascular bundles

122. Which of the following is not part of ground tissue system?
(1) Hypodermis (2) Pith
(3) Xylem (4) Endodermis

123. Stomatal apparatus does **not** include
(1) Stomatal pore
(2) Guard cells
(3) Surrounding subsidiary cells
(4) Hypodermal cells

124. Observe the following diagram.



Here, label (A), (B) and (C) represent

	(A)	(B)	(C)
(1)	Guard cells	Subsidiary cells	Epidermal cells
(2)	Subsidiary cells	Guard cells	Epidermal cells
(3)	Epidermal cells	Subsidiary cells	Guard cells
(4)	Subsidiary cells	Epidermal cells	Guard cells

(1) (1) (2) (2)
(3) (3) (4) (4)

125. Guard cells in grass leaf are
(1) Rectangular (2) Barrel shaped
(3) Kidney shaped (4) Dumb-bell shaped

126.



The above type of vascular bundles are found in
(1) Dicot leaf (2) Monocot root
(3) Monocot stem (4) Dicot stem

127. In dicot stem, the vascular bundles
(1) Are arranged in a ring (2) Are conjoint and closed
(3) Are radial and open (4) Are scattered

128. The endodermis is also known as starch sheath in
(1) Monocot stem (2) Dicot stem
(3) Monocot root (4) Dicot root

129. A waxy thick layer generally covers the epidermis which prevent water loss it is absent in
(1) Stem (2) Root
(3) Leaves (4) Flower

130. A tissue is a group of cells that are
(1) Different in origin
(2) Common in function
(3) Always similar in structure
(4) Only meristematic in activity

131. Epidermal tissue system does not include
(1) Subsidiary cells (2) Guard cells
(3) Conjunctive tissue (4) Trichomes

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132. Identify the **incorrect** statement about epidermis.
(1) Cells are parenchymatous with a small amount of cytoplasm
(2) Cuticle is absent in root epidermis
(3) The epidermal hairs on the stem are usually unicellular
(4) Leaf epidermis bears stomata

133. The innermost layer of the cortex rich in starch grains is
(1) Also referred to as casparian strips.
(2) Found in both stem and root of monocots.
(3) A part of stele.
(4) Found in the dicot stem.

134. Root differs from stem in having
(1) Endarch xylem
(2) Radial vascular bundle
(3) Collenchymatous ground tissue
(4) Absence of cortex

135. Vascular bundles of maize stem
(1) Have cambium between xylem and phloem
(2) Are scattered in ground tissue
(3) Are almost of uniform size
(4) Lack bundle sheath

ZOOLOGY

136. Osteoporosis, an age-related disease of skeletal system, may occur due to
(1) Immune disorder affecting neuromuscular junction leading to fatigue
(2) High concentration of Ca^{++} and Na^{+} in bones
(3) Decreased levels of estrogen
(4) Accumulation of uric acid leading to inflammation of joints

137. Choose the **correct** option to complete the analogy w.r.t. humans.
Joint between the carpal and metacarpal of thumb : Saddle joint :: Joint between the carpals : ___
(1) Saddle joint (2) Gliding joint
(3) Hinge joint (4) Ellipsoid joint

138. Ischium bone is a structural constituent of
(1) Pectoral girdle (2) Coxal bone
(3) Axial skeleton (4) Lower limbs

139. An auto-immune disorder of muscular system which affects the neuro-muscular junction eventually leading to paralysis of skeletal muscle is
(1) Myasthenia gravis (2) Muscular dystrophy
(3) Tetany (4) Osteoporosis

140. Functions of the skeletal system include all of the following, **except**
(1) Erythropoiesis (2) Locomotion
(3) Storage of minerals (4) Production of body heat

141. In an adult human, vertebrae of which category are fused?
(1) Sacral and coccygeal (2) Cervical and thoracic
(3) Thoracic and lumbar (4) Cervical and coccygeal

142. A is a large, triangular, flat bone situated in the dorsal part of the thorax between the second and the seventh ribs in humans.
Select the option that fills the above blank **correctly**.
(1) Coxal bone (2) Clavicle
(3) Scapula (4) Sternum

143. Select the **correct** match w.r.t. having the same number of bones in the human body.
(1) Cranial bones – Facial bones
(2) Carpals of one limb – Tarsals of one limb
(3) Cranial bones – Carpals of one wrist
(4) Vertebrae – Skull bones

144. In our body which bone does **not** articulate with any other bone directly?
(1) Hyoid (2) Maxillae
(3) Lacrimal (4) Palatine

145. Select the **odd one** w.r.t. the bones of axial skeleton.
(1) Ear ossicles (2) Cranium
(3) Sternum (4) Clavicle

146. Tetany can be caused/characterised by all of the following, **except**
(1) Low Ca^{+2} level in body fluids
(2) Wild contraction in muscles
(3) Due to hyposecretion of thyrocalcitonin
(4) Due to hyposecretion of parathyroid hormone

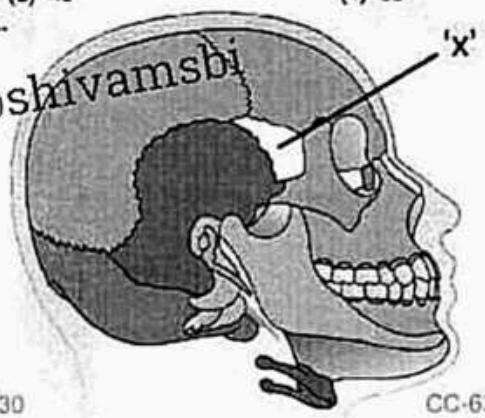
147. The type of joint shown by the flat skull bones which are fused end to end, is
(1) Fibrous joint (2) Cartilaginous joint
(3) Ball and socket joint (4) Pivot joint

148. At the fusion point of ilium, ischium and pubis, is a cavity to which _____ articulates.
Choose the **correct** option to fill in the blank.
(1) Acetabulum (2) Pubic symphysis
(3) Coxal bone (4) Femur

149. The articular surface of scapula with which the head of humerus interacts to form the shoulder joint is
(1) Glenoid cavity (2) Acromion process
(3) Clavicular notch (4) Acetabulum

150. Each limb of the human skeletal system is made up of how many bones?
(1) 25 (2) 30
(3) 40 (4) 35

151.



Choose the **incorrect** statement w.r.t. the bone marked as 'X'.
(1) It is unpaired
(2) It's a cranial bone
(3) It's a facial bone
(4) It is a part of axial skeleton

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152. Which of the following pairs of ribs do **not** articulate ventrally to sternum, but join the 7th rib with the help of hyaline cartilage?
(1) 1st to 5th (2) 8th, 9th and 10th
(3) 11th and 12th (4) 10th and 12th

153. Choose the **odd** one out w.r.t. bones of hindlimbs.
(1) Femur (2) Tarsals
(3) Metacarpals (4) Fibula

154. The bone that is present at base of buccal cavity is
(1) Hyoid bone (2) Vomer
(3) Maxilla (4) Frontal

155. In humans, the number of true ribs are equal to the number of
(1) Carpals (2) Facial bones
(3) False ribs (4) Floating ribs

156. Which of the following are included in PNS?
(a) Nerves
(b) Nerve fibres
(c) Ganglia
(d) Plexuses
Choose the **correct** option w.r.t. humans
(1) (a), (b), (c) and (d) (2) Only (b) and (c)
(3) Only (a) and (d) (4) Only (c) and (d)

157. A unipolar neuron
(1) Does not have soma or body
(2) Possesses a single dendrite and single axon process
(3) Does not have dendrites
(4) Has numerous axons

158. The organ system that provides an organised network of point to point connections for quick coordination through synapses is
(1) Endocrine system (2) Locomotory system
(3) Neural system (4) Circulatory system

159. Choose the correct option for neurons.
(1) They form the myelin sheath in CNS
(2) They transmit impulses always on sub-threshold stimulus
(3) They are not excitable cells
(4) They depolarise upon receiving threshold stimulus

160. Autonomic nervous system includes
(1) Parasympathetic and sympathetic nervous systems
(2) Parasympathetic and peripheral nervous systems
(3) Parasympathetic and central nervous systems
(4) Central nervous system and peripheral nervous system

161. The part of forebrain which lies at the base of the structure which is a major coordinating centre for sensory and motor signalling, contains centres that control
(1) Gastric secretions (2) Respiration
(3) Cardiovascular reflexes (4) Body temperature

162. During the resting state of a nerve fibre, the axonal membrane is
(1) More permeable to K^+ than Na^+
(2) More permeable to Na^+ than K^+
(3) Equally permeable to Na^+ and K^+
(4) More permeable to Cl^- and K^+

163. Which of the following is **not** a part of brain stem?
(1) Pons varolii (2) Cerebellum
(3) Midbrain (4) Medulla oblongata

164. Which of the following is present at the node of Ranvier in a neuron of the CNS?
(1) Myelin sheath (2) Axolemma
(3) Schwann cell (4) Oligodendrocyte

165. Electrical synapses
(1) Involve neurotransmitter molecules like dopamine
(2) Permit only unidirectional movement of impulses
(3) Allow nerve impulses to travel faster than seen in a chemical synapse
(4) Work through tight junctions that allow the passage of ions

166. Exclusive feature of nerve cells is the presence of
(1) Node of Ranvier and flagella
(2) Oligodendrocytes and cilia
(3) Dendrite and cyton
(4) Ribosomes and endoplasmic reticulum

167. The cerebral cortex have greyish appearance due to the presence of concentrated
(1) Myelinated axons
(2) Myelinated synaptic vesicles
(3) Neuron cell bodies
(4) Myelinated synaptic knobs

168. The structure which connects the left and right cerebral hemispheres in human brain is
(1) Corpus callosum (2) Cerebral aqueduct
(3) Corpora quadrigemina (4) A deep longitudinal cleft

169. Amygdala is a
(1) Component of limbic system
(2) Characteristic arrangement of grey and white matter in cerebellum
(3) Tree-like arrangement of neurons in cerebrum
(4) Component of mid-brain

170. Meninx which is in contact with the brain tissue and present in the form of a highly vascular thin layer is
(1) Dura mater (2) Arachnoid
(3) Pia mater (4) Coracoid

171. Which of the following organisms has a primitive nerve net but no brain?
(1) Earthworm (2) Spider
(3) *Hydra* (4) Cockroach

172. In brain, which area is responsible for complex functions like intersensory association, memory and communication?
(1) Association area
(2) Motor area
(3) Sensory area
(4) Dorsal root ganglionic area

173. All of the following centres are found in forebrain **except**
(1) Hunger (2) Thirst
(3) Pneumotaxic (4) Thermoregulatory

174. Limbic system is concerned with all of the following, except
(1) Regulation of sexual behaviour
(2) Expression of emotional reactions
(3) Motivation
(4) Regulation of breathing

175. Cerebral aqueduct is a canal which connects the cavity of diencephalon to cavity of medulla oblongata and passes through
(1) Cerebrum (2) Midbrain
(3) Cerebellum (4) Hypothalamus

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176. Complete the analogy by choosing the correct option w.r.t. humans.
Cerebral cortex : _____ neurons : : Embryonic stage : Unipolar neurons
(1) Multipolar (2) Bipolar
(3) Unipolar (4) Apolar

177. The part of the brain that is located between the thalamus/hypothalamus and pons is
(1) Cerebellum (2) Midbrain
(3) Medulla (4) Cerebrum

178. A drug addict died due to overdose of drugs which led to the depression of respiratory rhythm centre, present in
(1) Pons (2) Cerebellum
(3) Cerebrum (4) Medulla oblongata

179. In a neuron, synaptic knob possesses _____ which contain neurotransmitters.
Select the **correct** option to fill in the blank.
(1) Synaptic cleft (2) Synaptic vesicles
(3) Myelin sheath (4) Nodes of Ranvier

180. Read the following statements.
Statement (A): Synaptic cleft is an obligatory component of electrical as well as chemical synapses.
Statement (B): Transmission of an impulse across an electrical synapse is quite similar to impulse conduction along a single axon.
Select the **correct** option.
(1) Both statements (A) and (B) are incorrect
(2) Only statement (A) is correct
(3) Only statement (B) is correct
(4) Both statements (A) and (B) are correct

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